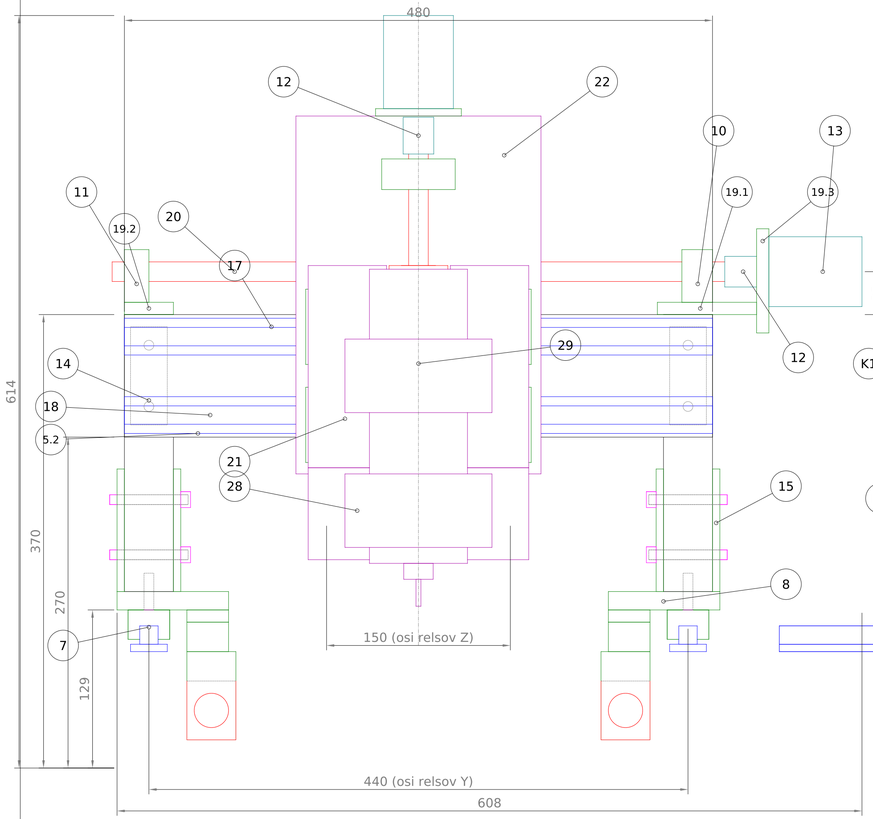
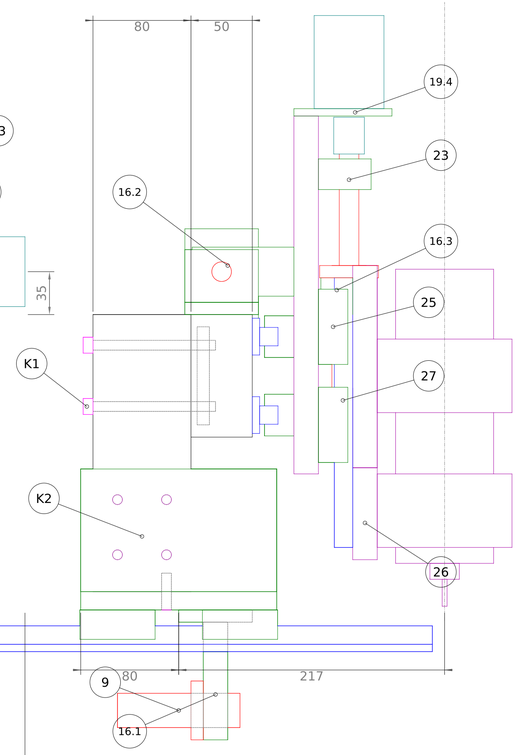
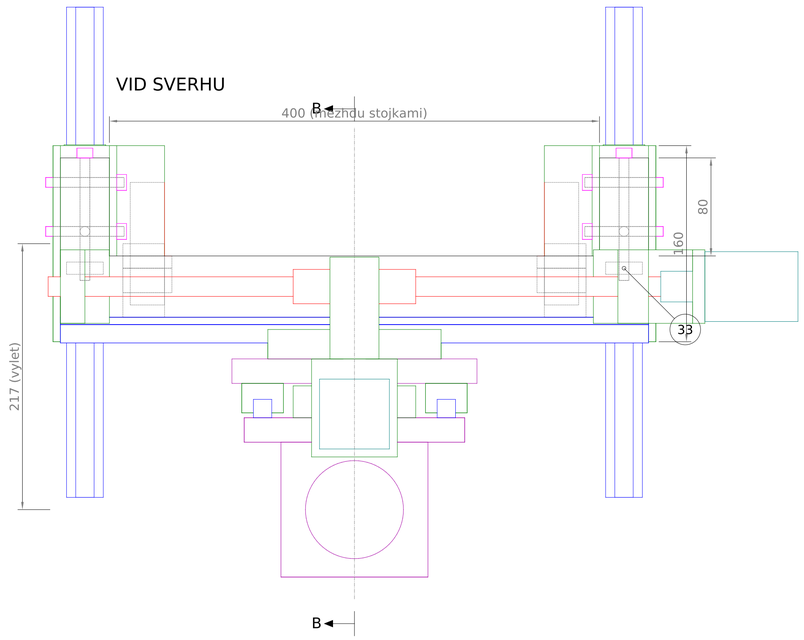
VID SPEREDI




VID SLEVA

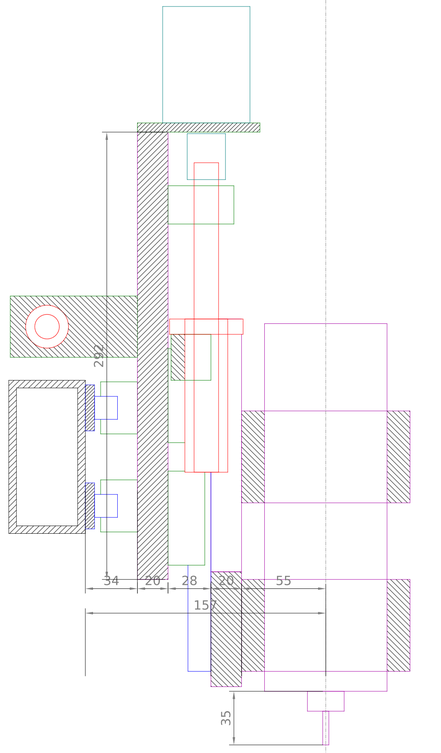



VID SVERHU




B-B (1:2)

stojki, plity 8 ne pokazany




kontsy instrumenta - na urovne stola (Z niz)

| Poz. | Naimenovanie | Razmer | Kol. | Primech. |
|---|---|---|---|---|
| 5.2 | Polosa X, stal 45 | 480x30x6 | 2 | list 2 |
| 7 | Karetka HGH15CA (os Y) | | 4 | pokup. |
| 8 | Plita opornaya, stal (svar. s 15) | 91x160x15 | 2 | 1 + 1 zerk. |
| 9 | Gajka SHVP Y DFU1605 (vint - chert. bazy) | | 2 | pokup. |
| 10 | Opora BK12 (X, Z) | | 2 | pokup. |
| 11 | Opora BF12 (X) | | 1 | pokup. |
| 12 | Mufta 6,35x10 (X, Z) | D25x30 | 2 | pokup. |
| 13 | Motor NEMA 23, 76 mm (X, Z) | | 2 | pokup. |
| 14 | Stojka, profil 40x80 paz 8 | L=226 | 2 | list 2 |
| 15 | Scheka, stal | 160x100x6 | 4 | list 3 |
| 16.1 | Kronshtejn gajki Y, svarnoj, stal | 40x60x106 | 2 | 1 + 1 zerk. |
| 16.2 | Kronshtejn gajki X, Al (kvadrat 40x40) | 40x83x40 | 1 | list 3 |
| 16.3 | Kronshtejn gajki Z, Al | 100x26x30 | 1 | list 3 |
| 17 | Balka X, truba Al 100x50x5 | L=480 | 1 | list 2 |
| 18 | Rels HGR15 (os X) | L=480 | 2 | pokup., E=30 |
| 19.1 | Plita BK12 X, Al | 81x60x10 | 1 | list 3 |
| 19.2 | Plita BF12 X, Al | 40x60x10 | 1 | list 3 |
| 19.3 | Plita motora X, Al | 60x85x10 | 1 | list 3 |
| 19.4 | Plita motora Z, stal | 70x80x6 | 1 | list 3 |
| 20 | SHVP DFU1605 C7 (os X) | L~520 | 1 | pokup. |
| 21 | Karetka HGH15CA (os X) | | 4 | pokup. |
| 22 | Support X, Al | 200x292x20 | 1 | list 2 |
| 23 | SHVP DFU1605 C7 (os Z) | L~205 | 1 | pokup. |
| 25 | Karetka HGH15CA (os Z) | | 4 | pokup. |
| 26 | Plita Z, Al | 180x240x20 | 1 | list 2 |
| 27 | Rels HGR15 (os Z) | L=220 | 2 | pokup., E=20 |
| 28 | Homut shpindelya D80, Al | 120x110x60 | 2 | list 3 |
| 29 | Shpindel 1,5 kVt ER11, D80 | | 1 | pokup. |
| 33 | Planka zakladnaya, stal | 80x30x10 | 2 | list 3 |
| K1 | Vint M8x100 DIN 912, 8.8 | | 4 | stojka - balka |
| K2 | Vint M8x65 DIN 912 + gajka DIN 985 | | 8 | scheki - stojka |
| K3 | Vint M8x30 DIN 912 | | 2 | plita 8 - stojka |
| K4 | Vint M4x12 DIN 912 | | 16 | karetki Y |
| K5 | Vint M4x16 DIN 912 | | 32 | karetki X, Z |
| K6 | Vint M4x16 DIN 912 | | 24 | relsy X, Z |
| K7 | Vint M5x12 DIN 7991 | | 24 | polosy, plity 19 |
| K8 | Vint M5x16 DIN 7991 | | 5 | plity 19.3, 19.4 |
| K9 | Vint M5 DIN 912, dlina po mestu | | 6 | BK12/BF12 |
| K10 | Vint M5x16 / M5x12 DIN 912 | | 4/4 | motory X / Z |
| K11 | Vint M5x16 DIN 912 | | 15 | flantsy gaek |
| K12 | Vint M6x25 DIN 912 | | 16 | kronshtejny 16 |
| K13 | Vint M6x35 DIN 912 | | 8 | homuty 28 |
| K14 | Vint M8x40 DIN 912 | | 2 | styazhka homutov |

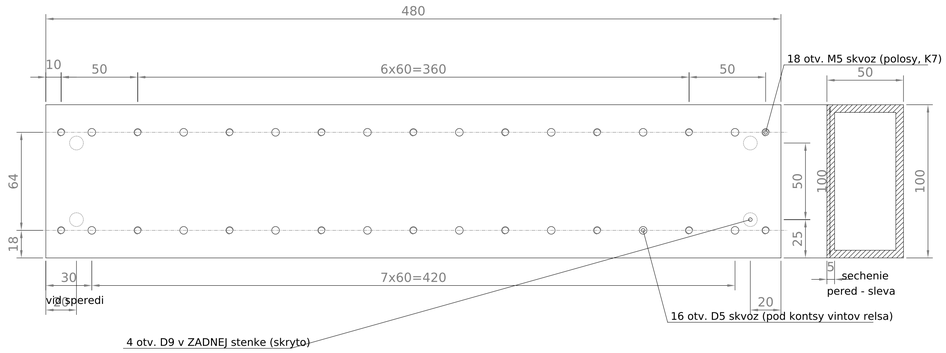
TEHNICHESKIE TREBOVANIYA

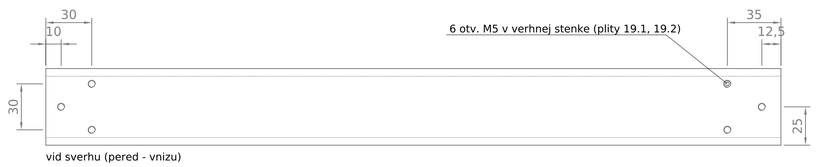
1. Razmery v mm. Polozhenie: instrument v tsentre polya, Z vnizu. Spec. v2.2. Baza so stolom - otd. chertezh.
2. Poryadok sborki:
   a) scheki 15 privarit k plite 8: snaruzhi sploshnoj shov, katet 6, kromka s razdelkoj 30x4; vnutri paza - tolko pered stojkoj (0...65 ot pereda), pod stojkoj NE varit; zatem frezerovat niz plity (ploskostnost 0,02) i paz 40 +0,1...+0,3 mezhdu schekami;
   b) plity 8 na karetki Y (K4) - DO ustanovki stoek: 4 vinta iz 8 okazhutsya pod stojkoj;
   c) stojki 14 v paz, K3 snizu; otv. D9 v stojke sverlit po schekam, K2 - 25 N*m;
   d) planki 33 zavesti v trubu s tortsov, balka 17 na stojki (K1): perpendikulyarnost relsam Y <= 0,05 na 300 (diagonali portala), K1 - 25 N*m;
   e) polosy 5.2 (K7): obschaya ploskost <= 0,02; bazovyj rels 18 - po linejke <= 0,01/100, vtoroj parallelno <= 0,02 po indikatoru s karetki; M4 - 4 N*m;
   f) support 22: SNACHALA karetki Z (K5 s tyla), POTOM karetki X (K5 speredi) - inache ne dostat;
   g) plita Z 26: relsy 27, homuty 28, kronshtejn 16.3; rels Z zavesti v karetki Z sverhu;
   h) BK/BF na plity 19, vint X vyverit: parallelnost relsam <= 0,05 v dvuh ploskostyah (folga pod plity); kronshtejny gaek 16 zatyagivat POSLEDNIMI, progonyaya karetku iz kraya v kraj; to zhe dlya Y i Z;
   i) shpindel: perpendikulyarnost stolu <= 0,05/100 v dvuh ploskostyah (folga pod homuty).
3. Rezba v Al: fiksator srednej prochnosti.
4. Razmery kupnyh (BK/BF, flanets DFU1605, motor, shpindel) sverit do sverlovki.

Portal s osyami X i Z. Sborka

| | | |
|---|---|---|
| Spec. v2.2. Detali - listy 2, 3 | Masshtab 1:2,5; 1:2 | List 1 / 3 |

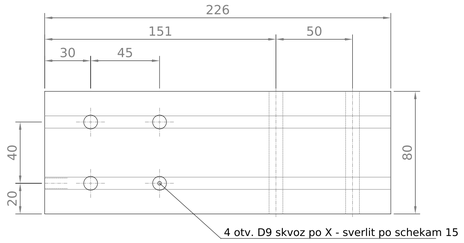
## Poz. 17  Balka X, truba Al 100x50x5 (6060/6063-T6)  1 sht.  M 1:2



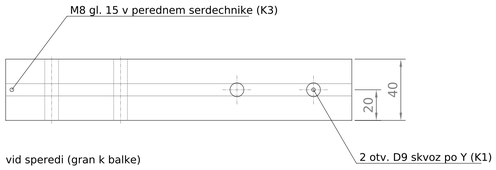




vid sverhu (pered - vnizu)

## Poz. 14  Stojka, profil 40x80 paz 8  2 sht.  M 1:2

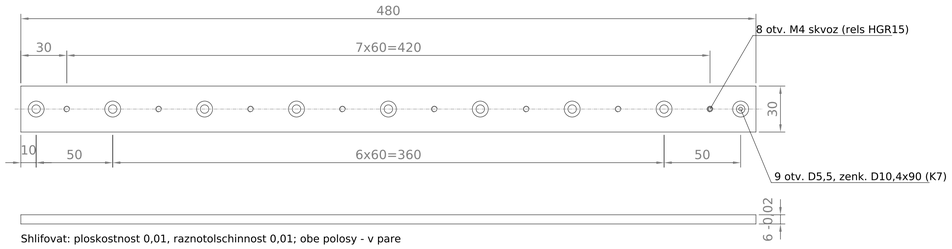



vid sboku (gran k scheke); niz - sleva, pered - vnizu




vid speredi (gran k balke)

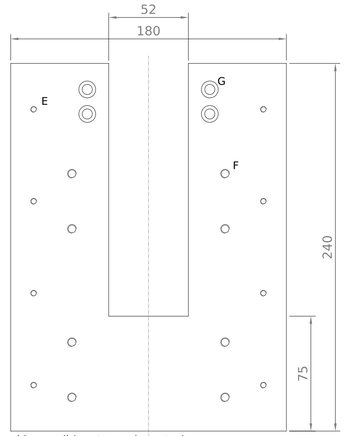
## Poz. 5.2  Polosa X 30x6, stal 45  2 sht.  M 1:2




Shlifovat: ploskostnost 0,01, raznotolschinnost 0,01; obe polosy - v pare

TEHNICHESKIE TREBOVANIYA K DETALYAM (LISTY 2, 3)
1. Razmery v mm, D - diametr. Neukazannye predelnye otkloneniya +-0,2; koordinaty otverstij +-0,1.
2. Plity 22, 26: litaya 5083 ili 6082-T6, obe storony frezerovat, ploskostnost 0,02, t = 20 +-0,05.
3. Truba 17: tortsy podrezat, perpendikulyarnost 0,1; otv. v perednej stenke sverlit po polose 5.2 kak po konduktoru.
4. Profil 14 - kak list 2 bazy: 6060/6063-T6, rezba M8 metchikom v serdechnike bez rassverlivaniya.
5. Plita 8 + scheki 15, kronshtejn 16.1 - St3 / S235; svarnye uzly obrabatyvat posle svarki (list 1, 2a).
6. Kromki pritupit. Pazy profilya - uslovno.

## Poz. 26  Plita Z, Al 6082-T6 / 5083  1 sht.  M 1:2



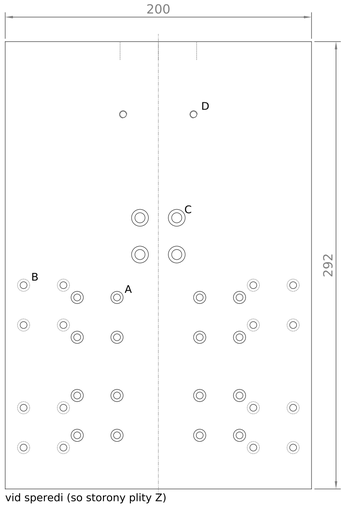

vid speredi (so storony homutov)
okno: R3 v uglah

Poz. 26: otverstiya. X - ot osi, Y - ot niza

| Obozn. | Otverstie | Kol. | X | Y |
|---|---|---|---|---|
| E | M4 skvoz (relsy Z, K6) | 8 | +-75 | 30; 90; 150; 210 |
| F | M6 skvoz (homuty, K13) | 8 | +-50 | 22; 58; 132; 168 |
| G | D6,6, tsek. D11 gl. 6,5 speredi | 4 | +-40 | 207; 223 |

G - kronshtejn 16.3 (K12), s tylnoj storony

## Poz. 22  Support X, Al 6082-T6 / 5083  1 sht.  M 1:2




vid speredi (so storony plity Z)

Poz. 22: otverstiya. X - ot osi, Y - ot niza

| Obozn. | Otverstie | Kol. | X | Y |
|---|---|---|---|---|
| A | D4,5, tsek. D8 gl. 8 speredi | 16 | +-27; +-53 | 35; 61; 99; 125 |
| B | D4,5, tsek. D8 gl. 8 SZADI | 16 | +-62; +-88 | 27; 53; 107; 133 |
| C | D6,6, tsek. D11 gl. 6,5 speredi | 4 | +-12 | 153; 177 |
| D | M5 gl. 12 speredi (BK12 Z) | 2 | +-23 | 244,5 |
| E | M5 gl. 12 v verhnem tortse | 3 | 0; +-25 | po t. 10 |

A - karetki X (K5), B - karetki Z (K5), C - kronshtejn 16.2 (K12)

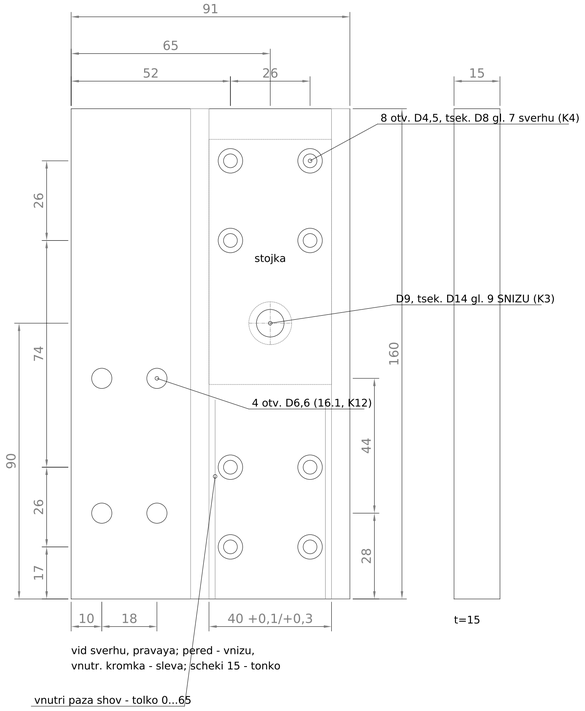
Portal. Detali

Spec. v2.2. Sborka - list 1

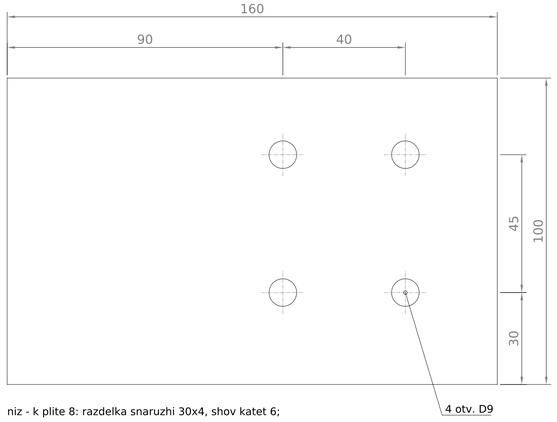
Masshtab 1:2

List 2 / 3

## Poz. 8 Plita opornaya, stal 15 1 + 1 zerk. M 1:1

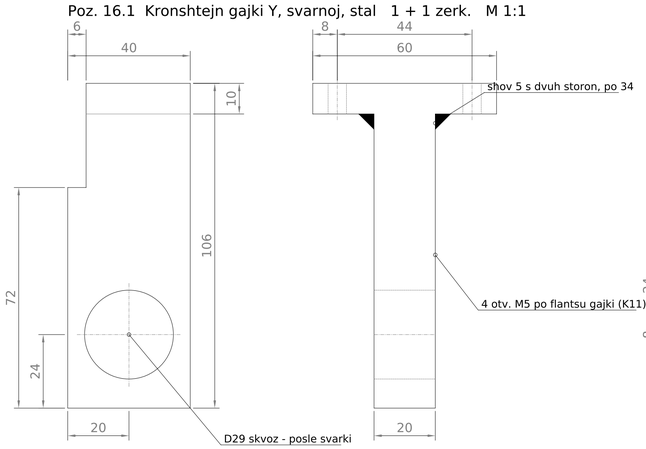



## Poz. 15 Scheka, stal 6 4 sht. M 1:1




niz - k plite 8: razdelka snaruzhi 30x4, shov katet 6;
pered - sleva; t = 6

## Poz. 16.1 Kronshtejn gajki Y, svarnoj, stal 1 + 1 zerk. M 1:1

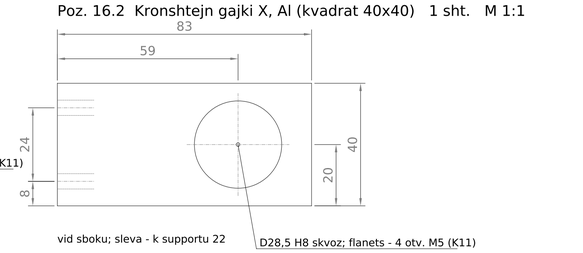



vid speredi

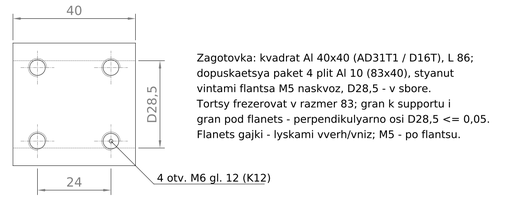
vid sboku (pered - sleva)

4 otv. M6 skvoz v polke (K12): X 16; 34, Y 8; 52.
Zagotovki St3: stojka - polosa 40x20, L=96 (stupenka 6x24 vypilit do svarki); polka - polosa 60x12, L=34.
Posle svarki frezerovat: verh polki v t=10, ploskostnost 0,05; zadnyuyu gran (pod flanets) - perpendikulyarno verhu 0,05/40, t >= 18; D29 - os perpendikulyarna zadnej grani; zatem rezby. Grunt/kraska, krome privalochnyh poverhnostej.

## Poz. 16.2 Kronshtejn gajki X, Al (kvadrat 40x40) 1 sht. M 1:1



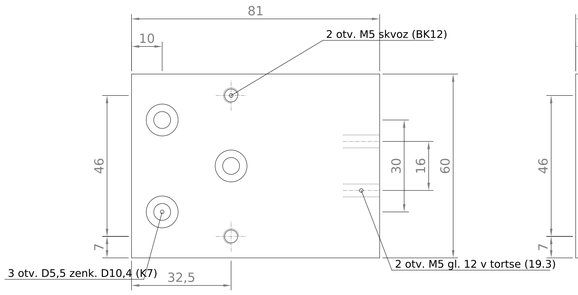

vid sboku; sleva - k supportu 22




Zagotovka: kvadrat Al 40x40 (AD31T1 / D16T), L 86;
dopuskaetsya paket 4 plit Al 10 (83x40), styanut vintami flantsa M5 naskvoz, D28,5 - v sbore.
Tortsy frezerovat v razmer 83; gran k supportu i gran pod flanets - perpendikulyarno osi D28,5 <= 0,05.
Flanets gajki - lyskami vverh/vniz; M5 - po flantsu.

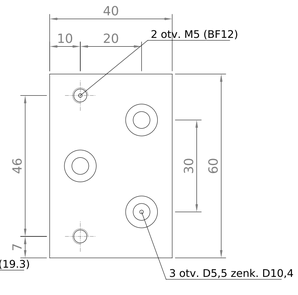
vid na gran k supportu

## Poz. 19.1 Plita BK12 X, Al 10 1 sht. M 1:1



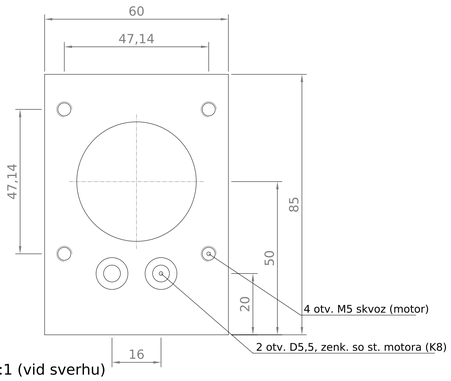

BK12 os - po u = 32,5; levyj kraj - k tsentru

## Poz. 19.2 Plita BF12 X, Al 10 1 sht.

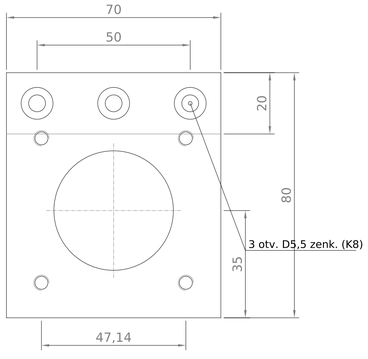



levyj kraj - torets balki

## Poz. 19.3 Plita motora X, Al 10 1 sht. (D39 H11 - poyasok motora D38,1)

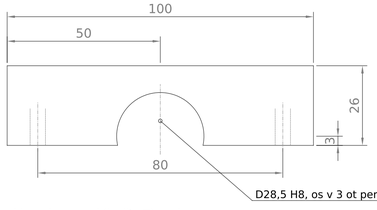



## Poz. 19.4 Plita motora Z, stal 6 1 sht.

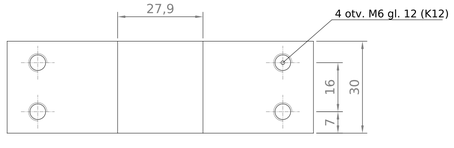



D39 H11; 4 otv. M5 (motor)
tonkaya - gran supporta 22

## Poz. 16.3 Kronshtejn gajki Z, Al 1 sht. M 1:1

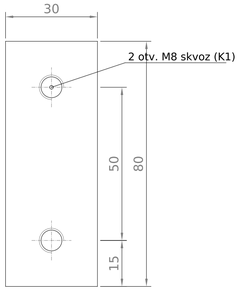



vid sverhu; pered (k plite Z) - vnizu

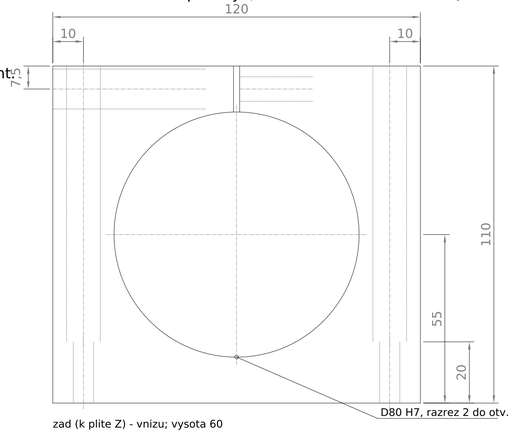



vid speredi; 3 otv. M5 sverhu - po flantsu (K11)

## Poz. 33 Planka zakladnaya, stal 10 2 sht.




## Poz. 28 Homut shpindelya, Al 6082-T6 2 sht. M 1:1 (vid sverhu)




zad (k plite Z) - vnizu; vysota 60
4 otv. D6,6, tsek. D11 do 20 ot zadnej grani, Z 12; 48 (K13)
styazhka: D9, tsek. D13 sleva / M8 sprava, Z 30 (K14)

| Portal. Detali | | |
|---|---|---|
| Spec. v2.2. Sborka - list 1 | Masshtab 1:1 | List 3 / 3 |